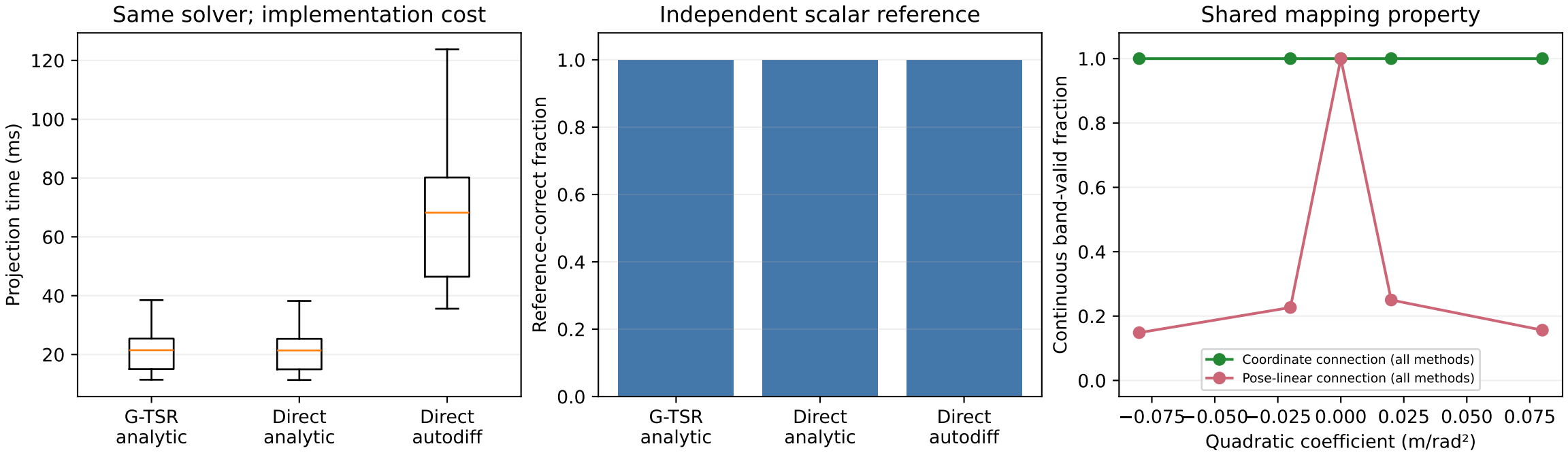
Same solver; implementation cost
Projection time (ms)
120
100
80
60
40
20
G-TSR analytic
Direct analytic
Direct autodiff
Independent scalar reference
Reference-correct fraction
1.0
0.8
0.6
0.4
0.2
0.0
G-TSR analytic
Direct analytic
Direct autodiff
Shared mapping property
Continuous band-valid fraction
1.0
0.8
0.6
0.4
0.2
0.0
Coordinate connection (all methods)
Pose-linear connection (all methods)
−0.075
−0.050
−0.025
0.000
0.025
0.050
0.075
Quadratic coefficient (m/rad²)